

A Comparison of Attention-Based and Attention-Free Object Detection Architectures for Motorcyclist Helmet-Use Detection

1st Author Name
Department Name
Institution Name
City, Country
email@example.com

2nd Author Name
Department Name
Institution Name
City, Country
email@example.com

Abstract—Helmet use is the primary protective factor for motorcyclists. Automatic compliance detection through traffic cameras can therefore support road-safety enforcement at scale. Attention mechanisms have become a dominant trend in computer vision, yet their benefit for helmet detection on small-scale datasets remains under-examined in controlled settings. This study compares four object-detection architectures that span the attention spectrum. The architectures are YOLOv8s as a pure CNN, YOLO11s as a CNN with partial attention through C2PSA blocks, RT-DETR-l as a transformer with full attention, and YOLOv8s+CBAM as a CNN augmented with a channel-spatial attention module. All models were trained on a public dataset of 1,803 images annotated for three classes, namely helmet, license plate, and motorcyclist. The training protocol was identical across models and included COCO transfer learning, 1280 pixel resolution, and matched augmentation. Each model was evaluated on a held-out test split with multiple seeds, and paired t-tests assessed significance. Under the unified protocol, no attention variant surpassed the plain YOLOv8s. The baseline reached the highest mAP@0.5 of 0.9592 ± 0.0014 , together with the fastest inference at approximately 296 FPS. Adding the CBAM module reduced mAP@0.5 by 0.0113 with statistical significance ($t(2) = 10.9$, $p = 0.008$, Cohen’s $d \approx 9.1$). YOLO11s and RT-DETR-l, by contrast, did not differ significantly from the baseline ($p > 0.05$). The baseline already approaches 0.96, which produces a ceiling effect, and the number of seeds is limited to two or three. The equivalence claims therefore represent an absence of evidence rather than evidence of absence. These findings hold under specific conditions, namely a single small dataset, a YOLO-tuned protocol, and 1280 pixel resolution. Within this scope, the added complexity of attention does not pay off, and a simple CNN remains the most efficient choice for real-time helmet detection.

Index Terms—helmet detection, object detection, attention mechanism, YOLO, RT-DETR, CBAM, motorcycle safety

I. INTRODUCTION

Motorcycles account for a large proportion of traffic-accident casualties, especially in developing countries where this mode dominates daily mobility. The helmet is the most decisive protective factor against fatal head injuries. Monitoring helmet-use compliance has therefore become an important target of road-safety policy. Manual observation does not scale to real traffic volumes because officers cannot watch every intersection at all times. Point-in-time surveys also produce a

biased picture of overall behaviour. Automatic object detection from camera footage offers a way out that can operate continuously and consistently [1].

Object detection has shifted over the past decade from accurate but slow two-stage approaches to fast single-stage detectors, with the YOLO family becoming the de facto standard for real-time applications [4], [19]. In parallel, attention mechanisms emerged from the Transformer [9] and were adapted to images through the Vision Transformer [10]. These mechanisms have transformed the architectural landscape of computer vision. Transformer-based detectors such as DETR [7] and its real-time descendant RT-DETR [6] now compete directly with CNNs. Lightweight attention modules such as Squeeze-and-Excitation [11] and CBAM [8] have also become a popular way to inject attention into existing CNN backbones.

The trend of adding attention to helmet-detection models has produced a number of positive reports. Several recent studies inject attention modules into YOLO-family detectors and report accuracy improvements on their respective helmet datasets. These reports consistently claim a benefit from attention. However, they generally compare a single model variant without rigorous experimental controls. Most lack repetition across seeds, a unified training protocol, and statistical significance testing. As a result, the reported performance differences could stem from hyperparameters, augmentation, or initialisation luck, rather than from the attention mechanism itself.

This trend gives rise to the implicit assumption that adding attention tends to improve performance. The assumption is reasonable on large-scale benchmarks such as COCO, where data diversity and volume give attention room to learn useful long-range dependencies. In the practical context of helmet detection, however, datasets are often small, curated, and limited in class count. Whether the additional complexity of attention pays off in such settings is therefore not yet clear.

This study addresses that gap with a single focused research question. **Can the improvements reported by attention-based studies on helmet detection be replicated in a controlled comparison that uses a unified protocol, rep-**

etition across seeds, and significance testing? The question positions the study as a structured test of the generalisability of the claimed benefits of attention in a controlled, small-scale setting.

The contributions of this paper are threefold. First, we present a controlled comparison of four detection architectures on the helmet-detection task, using identical hyperparameters, data, and augmentation. This design lets us attribute performance differences to architectural factors, particularly the presence and type of attention. Second, we conduct a multi-seed evaluation that reports the mean and standard deviation, paired t-tests, Cohen’s d effect size, and 95% confidence intervals. The conclusions therefore do not rest on the luck of a single initialisation. Third, we present a structured rebuttal of the assumption that adding attention always helps. On this dataset and under a YOLO-tuned protocol, no attention mechanism surpassed the pure CNN. The CBAM module even reduced accuracy in a statistically significant and consistent manner. We trace the cause of these patterns, discuss rival hypotheses, and summarise the implications for model-selection practice.

II. LITERATURE REVIEW

A. Helmet-Use Detection

Research on helmet detection has evolved from simple classification toward pipelines that can track motorcycles across frames and distinguish drivers from passengers. Siebert and Lin [1] demonstrated large-scale helmet-use detection from Myanmar traffic video using a deep-learning approach with a single-stage detector. They also released an annotated dataset that has become a community reference. Subsequent work has extended this direction with multi-task learning that registers helmet use per rider, so that pillion-rider cases can be handled and drivers can be distinguished from passengers.

Benchmark challenges such as the AI City Challenge 2023 Track 5 [2] have driven attention toward multi-class helmet-violation detection. The detailed class scheme distinguishes the helmet status of the driver and of the first and second passengers separately. This scheme exposes the problem of extreme class imbalance, since categories such as a helmetless second passenger appear very rarely. In the enforcement context, several approaches combine helmet detection with license-plate localisation and character recognition to identify violators [3]. Such pipelines are often supplemented by detection of more than two riders sharing a motorcycle.

B. Single-Stage Detectors and the YOLO Family

Single-stage detectors formulate detection as direct regression of bounding boxes and classes in a single network pass. They trade some accuracy for high speed [4], and the accuracy gap is partly narrowed by loss functions such as focal loss that address class imbalance in dense detection [12]. The YOLO family has become the backbone of real-time applications. Each iteration continually improves the backbone, the neck, the label-assignment strategy, and the augmentation pipeline [19]. YOLOv8 represents a mature CNN generation

without explicit attention modules. Its architecture relies on efficient convolutional blocks and a feature pyramid to handle multi-scale objects. YOLO11 introduces positional attention blocks called C2PSA into the feature path. This design makes YOLO11 an interesting midpoint on the attention spectrum, since it adds limited self-attention without fully switching to the transformer paradigm.

C. Attention Mechanisms in Computer Vision

Attention enables a model to selectively reweight information by highlighting relevant parts and suppressing irrelevant ones. Guo et al. [16] comprehensively surveyed attention mechanisms in computer vision and classified them into channel, spatial, temporal, and self-attention categories. Each category has its own strengths and limitations. The survey by Niu et al. [17] reviewed attention from a broader deep-learning perspective. They emphasised that the effectiveness of attention depends heavily on the volume of training data and on the complexity of the task.

The Transformer [9] introduced self-attention that models global dependencies among elements. The Vision Transformer [10] then proved this paradigm competitive for images when training data is sufficiently large. In detection, DETR [7] formulates detection as a set-prediction problem with a Transformer encoder-decoder. The architecture removes hand-crafted components such as anchor boxes and non-maximum suppression. The original DETR suffered from slow convergence and high computational cost. These weaknesses prompted RT-DETR [6], which adapts the idea into a real-time form through an efficient hybrid encoder. Another direction applies attention at the backbone level. The Swin Transformer [14] introduced an efficient hierarchical windowed attention as a detection backbone, and ViTDet [15] showed that a plain ViT backbone can also be used for detection. These backbone-level approaches generally excel when training data is abundant.

A different line of research injects attention into a CNN through lightweight modules without replacing the base architecture. Squeeze-and-Excitation [11] provides inter-channel attention by learning the importance weight of each feature channel. CBAM [8] extends this idea by combining channel attention and spatial attention sequentially. The model can therefore emphasise both what is important and where it is located. These modules are attractive because they are cheap, add few parameters, and are easy to attach to existing networks. Their effectiveness, however, depends on the task and on the scale of the data. On small datasets, a randomly initialised attention module must learn from a limited training signal.

For the task of small-object detection, Cheng et al. [18] surveyed large-scale approaches and noted that small objects pose a particular challenge. Items such as helmets and license plates within wide frames require multi-scale feature strategies, high resolution, and specially designed augmentation. The findings of this survey are directly relevant to our study. The helmet and license plate classes in the dataset used here are relatively small within the frame.

D. Attention in Helmet Detection

A growing body of literature integrates attention mechanisms into helmet-detection models. Common patterns include adding channel-spatial attention modules to YOLO-family detectors, inserting feature-fusion blocks together with attention layers, and adopting deformable or self-attention components in the detection head. These studies typically report improvements over their respective baselines on custom helmet datasets.

Although these reports consistently show a benefit from attention, a methodological pattern is worth noting. First, most studies report results from a single configuration without repetition across seeds. It is therefore unclear whether the reported improvement is consistent across initialisations. Second, the training protocol, including augmentation, resolution, number of epochs, and learning rate, is generally tuned separately for each model. The performance differences therefore reflect the combined effect of architecture and hyperparameters rather than attention alone. Third, no study reports statistical significance testing of the performance differences. This combined gap directly motivates our study, which is designed to isolate the contribution of the attention mechanism through a unified protocol, multi-seed repetition, and formal statistical testing.

E. Research Gap

The helmet-detection literature tends to report a single model configuration with varying protocols, often without repetition across seeds. The causal attribution to the attention mechanism is therefore weak. When a study reports that an attention-based model outperforms a baseline, it is difficult to be certain whether that advantage comes from attention or from differences in resolution, augmentation, number of epochs, or initialisation. The studies reviewed in Section II.D are no exception. They report a benefit from attention but do not include the experimental controls needed to attribute the cause firmly. This study fills that gap through a controlled, repeated comparison. We explicitly vary the presence and type of attention as the main axis, while holding all other factors constant. The held-constant factors include hyperparameters, augmentation, data, and input resolution.

III. METHODOLOGY

A. Dataset

We use a public helmet-detection dataset from Roboflow Universe, namely the NCKH 2023 Helmet Detection Project, version 19, released under the MIT license. The dataset contains 1,803 annotated images in YOLO format with three classes, namely helmet, license plate, and motorcyclist. A fixed data split was used throughout the experiments to ensure no re-randomisation across runs. The split comprises 1,563 training images, 140 validation images, and 100 test images. All final metrics were computed on the test split, which was never seen during training, to avoid data leakage. The annotations cover three classes that are directly relevant to the enforcement scenario. The rider class provides context, the helmet class is

the compliance object, and the license plate class serves as an identification anchor.

B. Architectures Compared

Four architectures were chosen so that they occupy different points on the spectrum of attention usage, as summarised in Table I. The selection is deliberate. It moves from a CNN with no attention at all, through a CNN with partial attention, to a transformer with full attention, and finally to a controlled attention intervention that merely adds a module to the baseline.

TABLE I
THE FOUR ARCHITECTURES AND THEIR POSITION ON THE ATTENTION SPECTRUM.

Model	Paradigm	Attention	Params
YOLOv8s	Single-stage CNN	None (baseline)	~11.17 M
YOLO11s	Single-stage CNN	Partial (C2PSA)	~9.4 M
RT-DETR-l	Transformer	Full (self-attention)	~32 M
YOLOv8s+CBAM	CNN + module	Channel and spatial	~11.51 M

The YOLOv8s+CBAM variant is a custom architecture that anchors the controlled attention experiment. Three CBAM modules were inserted at the outputs of the three detection scales, namely P3, P4, and P5, just before the detection head. The model is therefore identical to the plain YOLOv8s except for the addition of that attention. Placement at all three scales ensures that attention operates on small features at P3, where helmets and plates appear, on medium features at P4, and on large features at P5. The addition raises the parameter count from 11.17 million to 11.51 million, which is an increase of about three percent. The intervention is therefore lightweight and isolated, so any change in performance can be linked directly to the attention module.

It must be noted that RT-DETR-l, with about 32 million parameters, is far larger than the YOLO-s variants, which contain about 9 to 11 million parameters. The comparison therefore mixes the factors of architectural paradigm and model capacity. The choice of standard models is deliberate because it reflects the practical choices faced by practitioners. The limitation is acknowledged and discussed explicitly in Section VI.

C. Training Protocol

To ensure that performance differences can be attributed to the architecture, all models were trained with an identical protocol. We use the Ultralytics framework [5] on top of PyTorch and a single NVIDIA RTX 4090 GPU. Each model was initialised with transfer learning from COCO-pretrained weights [13]. For the CBAM variant, the attention layers were randomly initialised while the rest inherited COCO weights.

The uniform configuration included an input resolution of 1280 pixels, automatic batch-size determination according to GPU memory, and automatic optimiser selection. The augmentation was identical and included mosaic, HSV shifting,

and horizontal flipping. Early stopping was configured with a patience of 25 epochs out of a maximum of 100 epochs.

The high resolution of 1280 pixels was chosen because preliminary analysis showed that the helmet and license plate classes occupy small regions within the frame. A lower resolution would harm their detection. Reproducibility was enforced by fixing the seed for the `random`, `NumPy`, and `PyTorch` libraries, with `cuDNN` configured in deterministic mode. The configuration and the environment information, including library versions, seed, and device, were saved for each run. For statistical validity, each model was retrained with several seeds, namely 42, 0, and 1. The mean and standard deviation on the test split are reported. RT-DETR-1 was run with only two seeds because its training cost is far higher, namely about 190 minutes per run compared with about 26 minutes for the YOLO variants. This limitation is noted explicitly and is taken into account when interpreting variance.

D. Evaluation Metrics and Statistical Testing Protocol

The primary metrics are $\text{mAP}@0.5$ and $\text{mAP}@[.5:.95]$, following object-detection convention. They are supplemented by precision, recall, and inference speed in frames per second, denoted as FPS. FPS measurement in a multi-seed sweep is affected by the GPU load when sweeps run back to back. The reported FPS figures were therefore taken from a single run while the GPU was idle, with no other training or inference process running. The Ultralytics built-in benchmark was used on the full test split. The figures therefore reflect the true speed on the hardware configuration used. The distinction matters so that the speed comparison is not misleading due to measurement artefacts.

To assess the significance of accuracy differences, we paired the results between models on the same seed. We then applied a paired t-test to the $\text{mAP}@0.5$ difference of each seed pair, with a significance level of $\alpha = 0.05$. In addition to the p -value, we report the Cohen's d effect size and the 95% confidence interval for the mean difference. Pairing by seed controls for initialisation variation, so the test becomes more sensitive to the architectural effect. This protocol follows recent guidance on accounting for variance in machine learning benchmarks, which argues that single-run comparisons are unreliable and that variance from initialisation must be modelled explicitly [20]. The number of seeds is small, with $n = 3$ for the YOLO variants and $n = 2$ for RT-DETR-1. The test is therefore indicative. It can confirm consistent differences, but its power to reject equivalence is low. This limitation is discussed explicitly in Section VI.

IV. RESULTS

Table II summarises the performance of the four architectures on the test split. YOLOv8s achieved the highest $\text{mAP}@0.5$ together with the best inference speed. YOLO11s and RT-DETR-1 were comparable in accuracy but slower. YOLOv8s+CBAM was in fact the lowest on $\text{mAP}@0.5$.

TABLE II
PERFORMANCE COMPARISON ON THE TEST SPLIT (MEAN $\pm$ STANDARD DEVIATION).

Model	n	$\text{mAP}@0.5$	$\text{mAP}@[.5:.95]$	FPS
YOLOv8s	3	0.9592 $\pm$ 0.0014	0.6862 $\pm$ 0.0046	$\sim$ 296
YOLO11s	3	0.9569 $\pm$ 0.0046	0.6898 $\pm$ 0.0030	$\sim$ 189
RT-DETR-1	2	0.9588 $\pm$ 0.0151	0.6764 $\pm$ 0.0205	$\sim$ 55
YOLOv8s+CBAM	3	0.9479 $\pm$ 0.0004	0.6810 $\pm$ 0.0045	$\sim$ 290

A. Effect of the Attention Mechanism

The main axis of the study is the presence and type of attention. Under a training protocol arranged for YOLO, this axis shows no advantage from attention. Fig. 1 visualises the $\text{mAP}@0.5$ of the four models together with the standard deviation across seeds. YOLOv8s without attention leads.

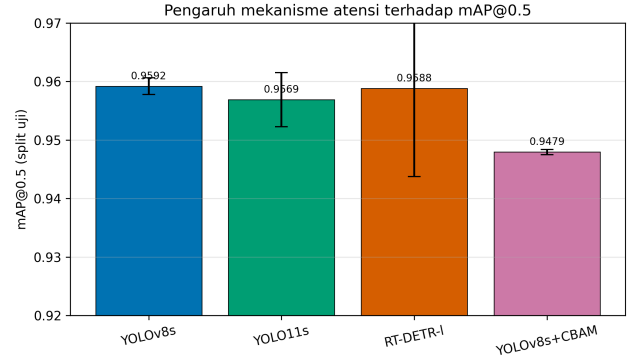


Fig. 1. Effect of the attention mechanism on $\text{mAP}@0.5$ on the test split. Error bars show the standard deviation across seeds. No attention-based variant surpasses YOLOv8s, and CBAM is in fact the lowest.

TABLE III
PAIRED T-TEST RESULTS FOR THE $\text{mAP}@0.5$ DIFFERENCE.

Comparison	Δ	$t(\text{df})$	p	d	95% CI
YOLOv8s vs YOLO11s	+0.0023	0.73 (2)	0.54	0.61	$[-0.011, +0.016]$
YOLOv8s vs RT-DETR-1	+0.0004	0.11 (1)	0.93	n/a	$[-0.140, +0.142]$
YOLOv8s vs CBAM	+0.0113	10.9 (2)	0.008	9.1	$[0.0068, 0.0157]$

Table III presents the paired t-test results for the $\text{mAP}@0.5$ difference between the baseline and each variant. The paired t-test confirms that YOLO11s does not differ significantly from the baseline ($\Delta = +0.0023$, $t(2) = 0.73$, $p = 0.54$, Cohen's $d = 0.61$, 95% CI $[-0.011, +0.016]$). RT-DETR-1 also does not differ significantly from the baseline ($\Delta = +0.0004$, $t(1) = 0.11$, $p = 0.93$, 95% CI $[-0.140, +0.142]$). The very wide confidence interval for RT-DETR-1 reflects an unstable estimate. The instability arises because only two seeds were available, and they produced far-apart results, namely 0.9482 and 0.9695. The equivalence of RT-DETR-1 is therefore better described as an absence of evidence of a difference rather than as evidence of equivalence. It must also be noted that the

Cohen’s d for YOLO11s versus the baseline is 0.61, which indicates a medium effect by convention. The effect does not reach significance because the statistical power is very low at $n = 3$.

In contrast, adding the CBAM module reduced mAP@0.5 by 0.0113, and this reduction is statistically significant. The paired t-test yields $t(2) = 10.9$, $p = 0.008$, Cohen’s $d \approx 9.1$, with a 95% confidence interval for the reduction of $[0.0068, 0.0157]$. The very large Cohen’s d indicates a substantial and consistent effect. The small standard deviation in both models, namely 0.0014 for the baseline and 0.0004 for the CBAM variant, confirms that the reduction is stable across the initialisations tested. We must emphasise that this result holds under a training protocol arranged for YOLO. The result shows that inserting CBAM is detrimental under those conditions, not that attention is universally detrimental.

B. Per-Class Performance

On the YOLOv8s baseline, the per-class accuracy follows a pattern that is consistent with the size and distinctiveness of the objects, as shown in Fig. 2. The motorcyclist class is the easiest to detect, with mAP@0.5 of 0.990, because the rider is large and prominent in the frame. The license plate class follows at 0.969, helped by its distinctive rectangular shape. The helmet class is the most challenging at 0.923. The difficulty arises because helmets are small, easily confused with other head accessories such as caps or hair, and varied in colour and viewing angle. This pattern confirms that the helmet class, which is in fact the most relevant for the application, is the main determinant of the room for improvement. Any future improvement effort should therefore focus on this class.

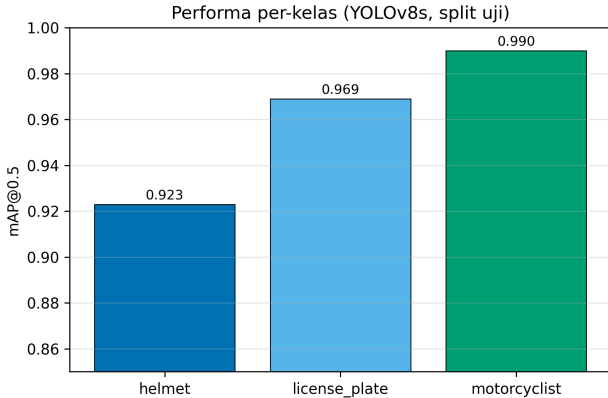


Fig. 2. Per-class performance of YOLOv8s on the test split. The helmet class is the most challenging because of its small size and visual ambiguity.

C. Speed Trade-Off

The speed difference is far larger than the accuracy difference, as shown in Fig. 3. RT-DETR-I runs at about 55 FPS, which is roughly five times slower than YOLOv8s at about 296 FPS. RT-DETR-I also demands a training time of about 190 minutes per run, compared with about 26 minutes

for the YOLO variants. YOLO11s sits in the middle at about 189 FPS. The CBAM variant runs at nearly the same speed as the baseline because the added module is lightweight. When the accuracy of the four models is practically equal, the computational cost of the transformer does not pay off on this task. Fig. 3 places YOLOv8s in the most advantageous corner because it leads on both axes simultaneously.

FPS was measured on a single run when the GPU was idle, with no other training or inference process running. The Ultralytics built-in benchmark was used on the full test split. The measurement was performed on the same hardware configuration for all models, namely an NVIDIA RTX 4090 GPU with batch size 1. The FPS figures are estimates at a single hardware point, and they will differ on other configurations.

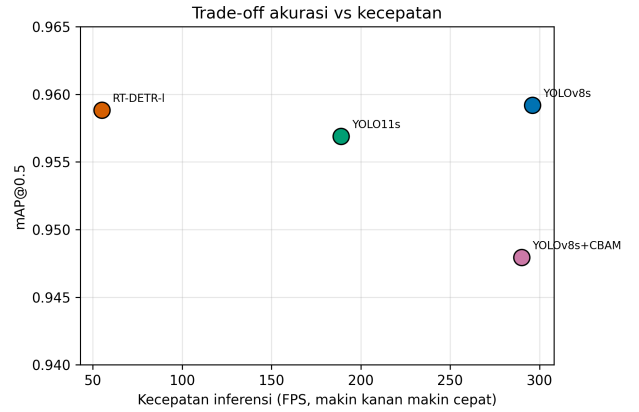


Fig. 3. Accuracy (mAP@0.5) versus speed (FPS) trade-off. YOLOv8s occupies the most advantageous position because it leads on both axes simultaneously.

D. Qualitative Analysis

Fig. 4 shows an example of YOLOv8s detection output on one test image. The model detects the rider, the helmet, and the license plate simultaneously with tight bounding boxes. This case illustrates the typical scene where all three classes are present together. Visual inspection of the test sample shows that the remaining errors tend to appear on very small or partially occluded helmets. The qualitative pattern is consistent with the quantitative finding that the helmet class is the most difficult.

V. DISCUSSION

The central finding of this study runs counter to the common intuition shaped by the attention-based helmet-detection literature reviewed in Section II.D. On a small-scale helmet-detection dataset, and under a training protocol arranged for YOLO, no attention mechanism surpassed the pure CNN. The CBAM module was even significantly detrimental. We discuss six complementary explanations that can account for this pattern.

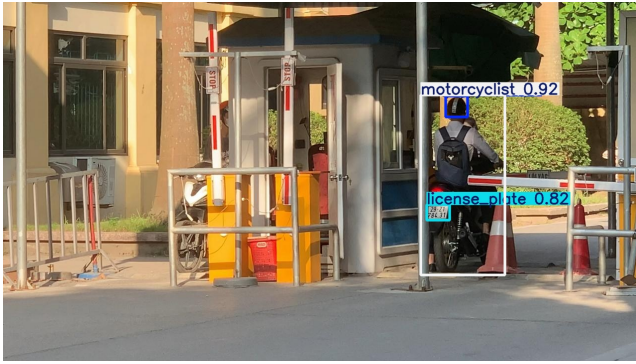


Fig. 4. Example of YOLOv8s detection output on a test image. The rider, the helmet, and the license plate are all detected simultaneously.

A. The Dataset Is Relatively Small and Easy

The training set contains only 1,563 images, and the baseline already reaches about 0.96 mAP@0.5. The room for improvement is therefore very narrow. When a simple model already approaches the upper bound achievable on this data, almost no gap remains for a more complex mechanism to demonstrate an advantage. Attention mechanisms are also data-hungry, especially transformer-based ones. They require large volume and diversity to learn useful dependency patterns [16], [17]. In the small-data regime, their theoretical advantage does not materialise. The model instead bears the burden of unused capacity.

B. Randomly Initialised Attention Layers Can Disrupt Pre-trained Features

In the CBAM variant, most weights are inherited from COCO. The attention module, however, starts from scratch. This untrained module inserts a transformation into the network. At the start of training, the transformation disrupts the mature feature flow obtained from transfer learning. With limited data, there is not enough gradient signal to recover from that disruption, let alone surpass the baseline. The result is a small but consistent reduction that we observe to be stable across seeds. The explanation is consistent with the general observation that additional modules are most beneficial when trained jointly from the start on large data. Their benefit is reduced when they are attached to a pretrained network with limited data.

C. Complexity Is Not Free

Attention adds parameters and computation. When there is no accuracy gain, this addition becomes pure cost. The consequence is most evident in RT-DETR-l, which is five times slower while showing no mAP reward. The same logic applies conceptually to CBAM, which adds a computational path at every detection scale. In real-time settings where the computational budget is limited, this cost translates into fewer frames processed per second.

D. The Cost of Errors Is Asymmetric in Safety Contexts

In safety-enforcement applications, the consequence of a false negative is far more serious than that of a false positive. Failing to detect a helmetless rider lets a violator escape. Flagging a helmeted rider as a violator merely creates a verifiable mistake. On the YOLOv8s baseline, the mAP@0.5 for the helmet class is 0.923, which means that about 7.7% of helmets are not correctly detected. In the context of automatic enforcement, this implies that nearly one in thirteen violators escapes undetected. Increasing recall for the helmet class, even at the expense of precision, may therefore be valued more in real applications than overall mAP. The design of a loss function that weights the helmet class more heavily, or the use of metrics that explicitly account for the asymmetric cost such as a weighted F1 score, is worth exploring for deployments aimed at enforcing safety.

E. The Gap Toward Real Deployment

The dataset used here is relatively clean. It has good lighting, minimal occlusion, and low object density. Real road CCTV conditions are very different. They include heavy occlusion by other riders, nighttime lighting, rain, and high traffic density. Under these challenging conditions, the capacity of attention to model global context may become useful. Cheng et al. [18] stress that small objects in dense frames require more sophisticated feature strategies, including multi-scale fusion and high input resolution. Under more challenging conditions, attention mechanisms may therefore provide benefits that are not observed on the relatively easy dataset used in this study.

F. A Rival Hypothesis for the CBAM Reduction

The mAP@0.5 reduction caused by CBAM does not automatically mean that CBAM is useless for helmet detection. Several alternative design factors can explain the reduction. First, the module placement may be suboptimal. We inserted CBAM only at the detection head, namely P3, P4, and P5, rather than in the backbone. Placement in the backbone, as proposed in some prior helmet-detection work, might yield different results. Second, the fresh layers may disrupt the COCO-pretrained weights. The available data may be insufficient to recover from this disruption. Third, the number of training epochs may not be enough for the new module to converge. Fourth, the learning rate was not tuned separately for the attention-module parameters. A systematic ablation that varies placement, trains from scratch, uses more epochs, and applies a separate learning rate is needed to isolate the exact cause. Such an ablation is part of future work.

G. Placement in the Broader Literature

These findings can be placed in the context of the broader literature. The advantages of transformer architectures and attention modules are most consistently reported on large-scale benchmarks with high diversity [16], [17]. On specialised tasks with limited data, several studies find that a well-arranged CNN remains competitive or superior. Our results

add evidence in that direction, specifically for the helmet-detection domain. They also underscore that architecture selection should be guided by the characteristics of the data and the computational budget. It should not be guided by architectural trends alone.

It is important to delimit this claim to avoid overgeneralisation. This study does **not** conclude that attention is useless in general. The narrower claim is as follows. Under a single training protocol arranged for YOLO, on a single small and relatively easy dataset, and with a baseline that already approaches the ceiling, adding attention did not improve accuracy. In the case of CBAM, it even reduced accuracy in a statistically significant manner. The equivalence of YOLO11s and RT-DETR-l should be read as an absence of evidence of a difference rather than as evidence of equivalence. The ceiling effect suppresses statistical power, so a small benefit from attention, if any exists, is hard to detect. The plausible rival hypothesis that the attention-based models have not been trained on their optimal recipe also cannot be ruled out. That hypothesis becomes part of the agenda for future work. In other words, this finding is most aptly read as a practical caution against the assumption that adding attention is sure to help. It is not a verdict against attention mechanisms in general.

The practical implication, within that scope, is clear. For real-time helmet detection on similar data, a simple CNN architecture such as YOLOv8s is the most rational choice. It provides the highest accuracy, the lightest footprint, and the fastest inference. The finding also serves as a methodological reminder that claims of architectural superiority must be tested under controlled and repeated conditions before being adopted. Without repetition across seeds, small differences such as those observed here are easily misinterpreted as real advantages. They may simply lie within the range of random variation.

VI. THREATS TO VALIDITY AND LIMITATIONS

This study has a number of limitations that constrain the generalisation of the findings. Each limitation is described explicitly below.

A. Evaluation Limited to a Single, Relatively Clean Dataset

Generalisation to real conditions has not been tested. Real conditions include road CCTV footage with occlusion, fast motion, poor lighting, and high density. On more challenging data, the ability of attention to model context may in fact become useful.

B. RT-DETR-l Was Run on Only Two Seeds

The computational cost of RT-DETR-l is far higher than that of the YOLO variants, namely about 190 minutes per run. RT-DETR-l was therefore evaluated on only two seeds. The variance estimate is consequently less precise. The standard deviation of 0.0151 for RT-DETR-l contrasts with 0.0014 to 0.0046 for the YOLO variants. The 95% confidence interval is also very wide at $[-0.140, +0.142]$. The equivalence conclusion for RT-DETR-l therefore cannot be considered with the

same confidence as the comparisons among YOLO variants that use three seeds. The RT-DETR-l equivalence claim must have its level of certainty lowered.

C. The Comparison Is Not Size-Matched

RT-DETR-l contains about 32 million parameters, while the YOLO-s variants contain about 9 to 11 million. The comparison reflects the standard model choices faced by practitioners. It nevertheless mixes the factors of architectural paradigm and model capacity. A comparison that equalises the parameter count would provide a purer picture. Evaluating YOLOv8m, with about 25 million parameters, as a size-matched control for RT-DETR-l would isolate the paradigm effect from the capacity effect.

D. Hyperparameter Confounding

Hyperparameters were unified and not tuned specifically for each architecture. The protocol favours YOLO, which is the more mature architecture and the more sensitive to the default Ultralytics settings. Transformers and attention-based models may require a different training recipe. Such a recipe could include a longer warm-up, a different learning rate, or a special decay schedule. Without per-architecture tuning, the results of the attention-based models may represent a lower bound rather than their true potential.

E. The Ceiling Effect Suppresses Statistical Power

The baseline already reaches about 0.96 mAP@0.5. The room for improvement is therefore very narrow, and the number of seeds is small at two or three. The statistical power of the t-test is consequently very low. A concrete view helps. With only 100 test images, a difference of one or two mislabelled annotations can already shift mAP@0.5 by about 0.01. The 0.002 to 0.004 difference observed between YOLOv8s and YOLO11s therefore lies within a range of variation that could be caused by a discrepancy on one or two images. The equivalence claims for YOLO11s and RT-DETR-l must therefore be read as an absence of evidence of a difference, not as evidence of equivalence. Concluding that attention does not help in absolute terms would be a leap from absence of evidence to evidence of absence.

F. FPS Measured on a Single Configuration

The speed figures were taken from a single run to mitigate inter-run interference. They remain estimates on a single hardware configuration, namely an NVIDIA RTX 4090 GPU with batch size 1. The FPS results will differ on other devices. They also do not reflect end-to-end latency that would include pre-processing and post-processing.

G. The CBAM Reduction May Be an Artefact of the Experimental Design

The accuracy reduction caused by CBAM cannot be automatically attributed to the ineffectiveness of the attention module itself. Several alternative design factors can explain the reduction. First, the module placement may be suboptimal because CBAM was inserted only at the detection head and

not in the backbone. Second, the fresh layers may disrupt the COCO-pretrained weights. Third, the number of epochs may be insufficient for the new module to converge. Fourth, the learning rate was not tuned separately for the attention-module parameters. A systematic ablation that varies placement, trains from scratch, uses more epochs, and applies a separate learning rate is needed to isolate the exact cause.

H. A Single Dataset Without Difficulty Variation

Using a single dataset with relatively easy conditions limits the ability to test whether attention provides benefits under more challenging conditions. The conditions of good lighting and minimal occlusion are not representative of real CCTV. Generalisation to scenes with heavy occlusion, nighttime lighting, rain, and high traffic density has not been verified.

I. The Overall Number of Seeds Is Limited

Three seeds are sufficient to detect large effects such as the CBAM reduction with $d \approx 9.1$. The statistical power to detect small effects, however, is very limited. To confirm equivalence with an adequate level of confidence, a larger number of seeds is required. More sophisticated inference methods, such as Bayesian equivalence testing, could also help.

J. No Evaluation on a Size-Matched Attention-Based Architecture

A comparison with YOLOv8m, with about 25 million parameters, or with other medium-sized YOLO variants as a size-matched control for RT-DETR-l has not been carried out. Without this comparison, we cannot ascertain whether the equivalent RT-DETR-l result reflects the ineffectiveness of the transformer paradigm or merely a model-scale mismatch.

VII. CONCLUSION AND FUTURE WORK

We compared four object-detection architectures that occupy a spectrum of attention usage for motorcyclist helmet-use detection. The comparison used an identical training and evaluation protocol with multi-seed repetition. Under a YOLO-tuned protocol on a single, small, and relatively easy dataset, no attention mechanism surpassed the pure-CNN YOLOv8s. The attention mechanisms tested include partial attention through C2PSA, full attention through a transformer, and the CBAM module.

YOLOv8s delivered the highest accuracy at 0.9592 ± 0.0014 mAP@0.5, together with the best speed of about 296 FPS. Adding CBAM reduced mAP@0.5 by 0.0113 in a statistically significant manner ($t(2) = 10.9$, $p = 0.008$, Cohen's $d \approx 9.1$). The equivalence of YOLO11s and RT-DETR-l, with $p = 0.54$ and $p = 0.93$ respectively, is interpreted as an absence of evidence of a difference rather than as evidence of equivalence. The ceiling effect and the limited number of seeds explain this interpretation.

These findings hold under the specific conditions tested. The conditions are a single, small, and relatively easy dataset, a training protocol arranged for YOLO, and a resolution of 1280 pixels. Generalisation to other conditions requires further

validation. Such conditions would include larger and more challenging datasets, per-architecture-tuned hyperparameters, and architectures with comparable capacity. Within this scope, the finding is best read as a practical caution against the assumption that adding attention always helps on small-scale helmet-detection tasks. It is not a general verdict on attention mechanisms.

The recommendations for future work follow directly from the limitations above. First, the generalisation should be tested on larger, more diverse, and more challenging datasets. These should include real CCTV footage under difficult conditions such as occlusion, nighttime lighting, rain, and high density. Second, attention should be evaluated in a large-data regime where its advantage has a chance to emerge. Training from scratch should also be considered, rather than merely attaching modules to a pretrained network. Third, size-matched comparisons should be added. YOLOv8m, with about 25 million parameters, is a natural capacity control for RT-DETR-l. Per-architecture hyperparameter tuning should also be performed so that each model is tested under its best conditions. Fourth, a systematic CBAM ablation is needed. The ablation should vary placement across backbone, neck, and detection head. It should also include training from scratch, longer schedules, and a separate learning rate for the attention parameters. Fifth, a two-stage approach is worth exploring. Such an approach detects motorcycles first and then classifies helmets within the detected motorcycle regions. License plate detection can be integrated for enforcement scenarios, and a loss function that weights the helmet class more heavily can address the asymmetric error cost. Sixth, the number of seeds should be increased to at least five. Bayesian inference methods can also be used to strengthen equivalence or difference claims.

STATEMENTS

Data Availability. The dataset is publicly available through Roboflow Universe under the name NCKH 2023 Helmet Detection Project, version 19, with the MIT license. The training scripts, configurations, and experiment notebooks are available in the authors' research repository.

Conflict of Interest. The authors declare no conflict of interest.

Funding. No specific funding is reported for this study.

ACKNOWLEDGMENT

The authors thank the course instructors for their guidance throughout the project. The authors also acknowledge the use of AI-based tools for assistance in experiment design, analysis, and manuscript drafting. All methodological decisions, verification of numerical results, and final responsibility for the content rest with the authors.

REFERENCES

- [1] F. W. Siebert and H. Lin, "Detecting motorcycle helmet use with deep learning," *Accident Analysis & Prevention*, vol. 134, p. 105319, 2020.
- [2] M. Naphade et al., "The 7th AI City Challenge," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2023.

- [3] W. Jia et al., “Real-time automatic helmet detection of motorcyclists in urban traffic using improved YOLOv5 detector,” *IET Image Processing*, vol. 15, no. 14, pp. 3623–3637, 2021.
- [4] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, “You only look once: Unified, real-time object detection,” in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 779–788.
- [5] G. Jocher, A. Chaurasia, and J. Qiu, “Ultralytics YOLO,” 2023. [Online]. Available: <https://github.com/ultralytics/ultralytics>
- [6] Y. Zhao et al., “DETRs beat YOLOs on real-time object detection,” in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2024.
- [7] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, “End-to-end object detection with transformers,” in *Proc. European Conf. Computer Vision (ECCV)*, 2020, pp. 213–229.
- [8] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, “CBAM: Convolutional block attention module,” in *Proc. European Conf. Computer Vision (ECCV)*, 2018, pp. 3–19.
- [9] A. Vaswani et al., “Attention is all you need,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017, pp. 5998–6008.
- [10] A. Dosovitskiy et al., “An image is worth 16x16 words: Transformers for image recognition at scale,” in *Proc. Int. Conf. Learning Representations (ICLR)*, 2021.
- [11] J. Hu, L. Shen, and G. Sun, “Squeeze-and-excitation networks,” in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2018, pp. 7132–7141.
- [12] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, “Focal loss for dense object detection,” in *Proc. IEEE Int. Conf. Computer Vision (ICCV)*, 2017, pp. 2980–2988.
- [13] T.-Y. Lin et al., “Microsoft COCO: Common objects in context,” in *Proc. European Conf. Computer Vision (ECCV)*, 2014, pp. 740–755.
- [14] Z. Liu et al., “Swin Transformer: Hierarchical vision transformer using shifted windows,” in *Proc. IEEE/CVF Int. Conf. Computer Vision (ICCV)*, 2021, pp. 10012–10022.
- [15] Y. Li, H. Mao, R. Girshick, and K. He, “Exploring plain vision transformer backbones for object detection,” in *Proc. European Conf. Computer Vision (ECCV)*, 2022, pp. 280–296.
- [16] M.-H. Guo, T.-X. Xu, J.-J. Liu, Z.-N. Liu, P.-T. Jiang, T.-J. Mu, S.-H. Zhang, R. R. Martin, M.-M. Cheng, and S.-M. Hu, “Attention mechanisms in computer vision: A survey,” *Computational Visual Media*, vol. 8, no. 3, pp. 331–368, 2022.
- [17] Z. Niu, G. Zhong, and H. Yu, “A review on the attention mechanism of deep learning,” *Neurocomputing*, vol. 452, pp. 48–62, 2021.
- [18] G. Cheng, X. Yuan, X. Yao, K. Yan, Q. Zeng, and J. Han, “Towards large-scale small object detection: Survey and benchmarks,” *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 1, pp. 521–539, 2024.
- [19] J. Terven, D.-M. Córdova-Esparza, and J.-A. Romero-González, “A comprehensive review of YOLO architectures in computer vision: From YOLOv1 to YOLOv8 and YOLO-NAS,” *Machine Learning and Knowledge Extraction*, vol. 5, no. 4, pp. 1680–1716, 2023.
- [20] X. Bouthillier et al., “Accounting for variance in machine learning benchmarks,” in *Proc. Machine Learning and Systems (MLSys)*, vol. 3, 2021.